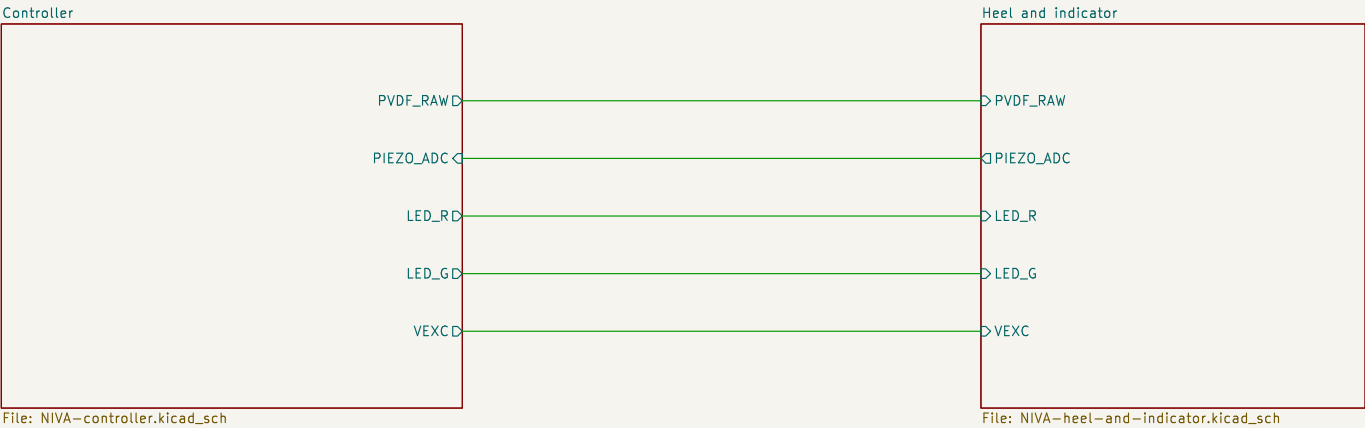


NIVA pod Rev C – one electrically connected hierarchy



Rails +3V3, GND and VBAT are global power symbols shared by both sheets.
Cross-sheet signals travel only through the four sheet pins drawn above.
ENGINEERING PROTOTYPE – NOT FOR FABRICATION. See README for the open review items.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.
NIVA

Sheet: /
File: NIVA-pod.kicad_sch

Title: NIVA pod | Rev C hierarchy

Size: A4 Date: 2026-09-28

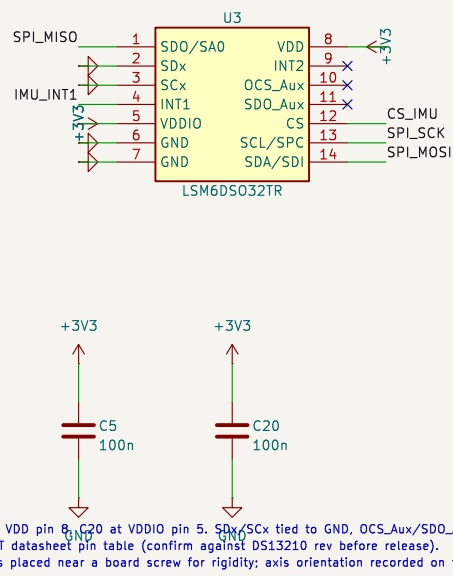
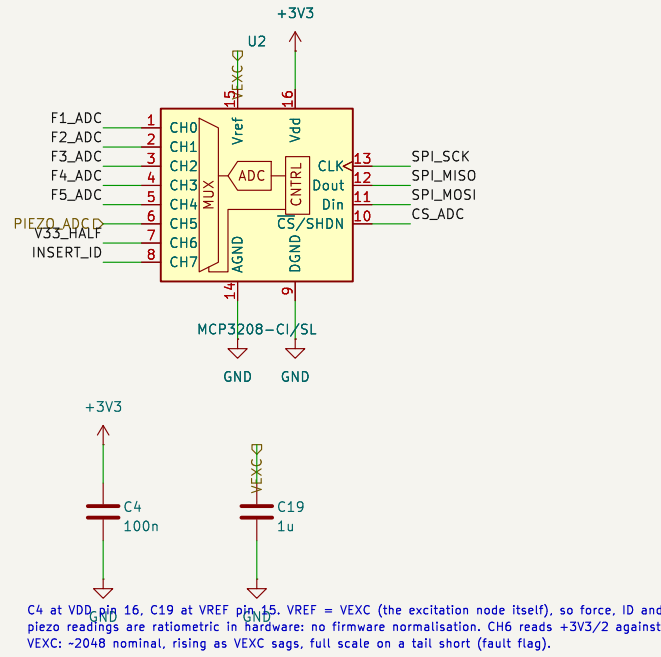
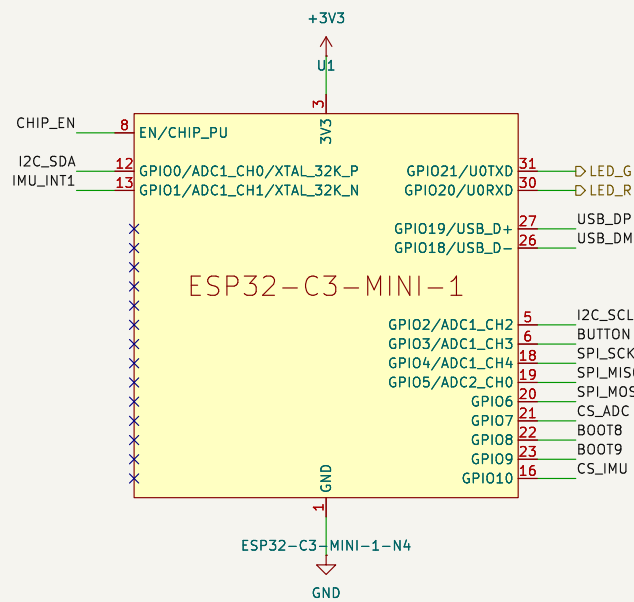
KiCad E.D.A. 9.0.9

Rev: C – ENGINEERING PROTO

Id: 1/3

01 ESP32–C3 MODULE, BOOT STRAPS, SERVICE

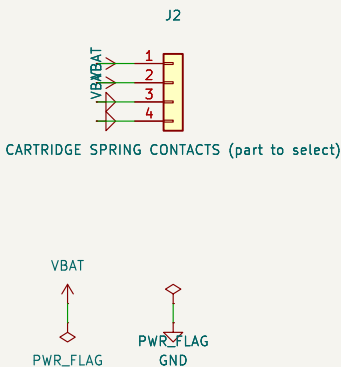
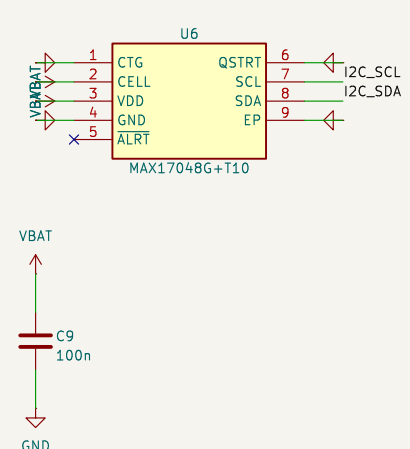
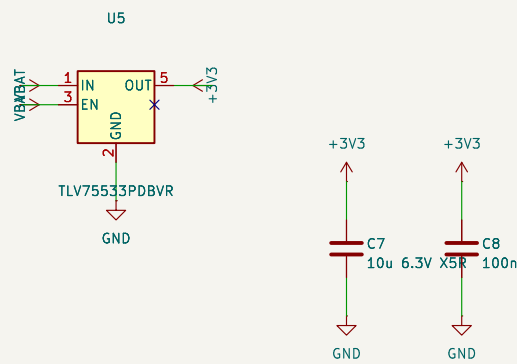
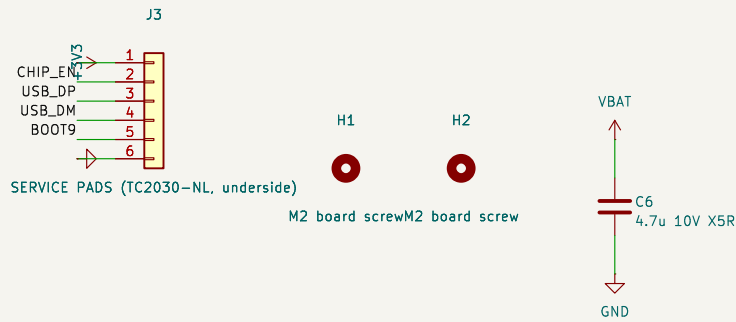
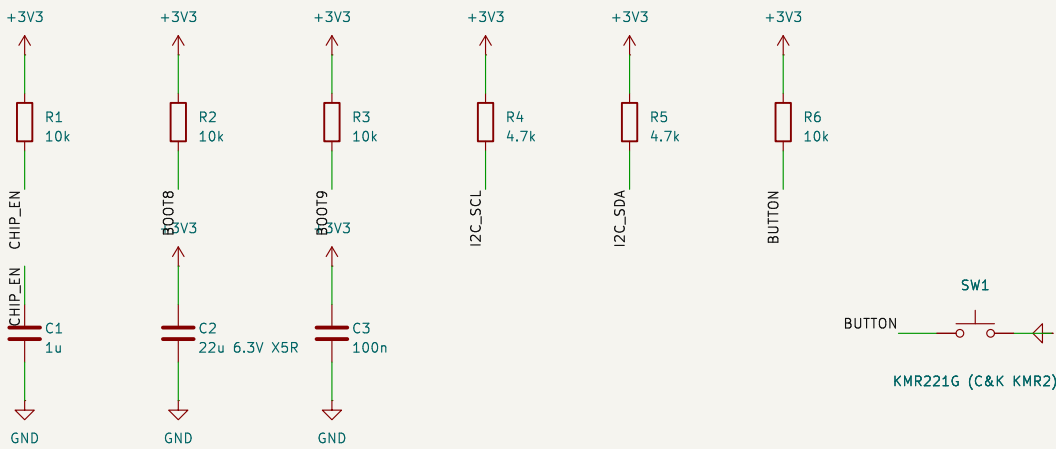
02 12–BIT ADC + INERTIAL SENSOR (SHARED SPI)



C4 at VDD pin 16, C19 at VREF pin 15. VREF = VEXC (the excitation node itself), so force, ID and piezo readings are ratiometric in hardware: no firmware normalisation. C16 reads +3V3/2 against VEXC: ~204B nominal, rising as VEXC sags, full scale on a tall short (fault flag).

C5 at VDD pin 8, C20 at VDDIO pin 5. SCL/SCx tied to GND, DCS_Aux/SDO_Aux and INT2 open, per ST datasheet pin table (confirm against DS3210 rev before release). IMU is placed near a board screw for rigidity; axis orientation recorded on the PCB silk.

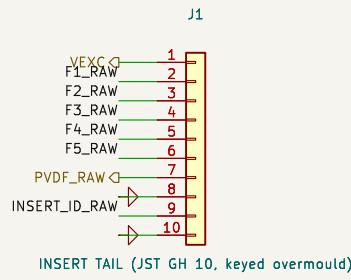
03 POWER FROM REMOVABLE CARTRIDGE (NO CHARGER ON THIS BOARD)



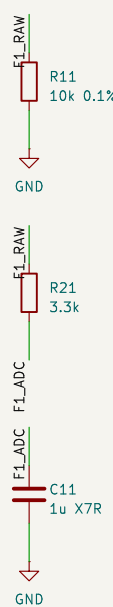
GPIO map (Rev C review of Rev B): I00 I2C_SDA, I02 I2C_SCL (strap: 4.7k pull-up keeps it high at reset), I03 BUTTON, I01 IMU_INT1, I04/5/6 SPI_SCK/MISO/MOSI, I07 CS_ADC, I010 CS_IMU, I020 LED_R, I021 LED_G, I018/19 USB (service only), I08 = BOOT strap (10k pull-up, pulled low only by the service jig), I09 = strap pull-up; otherwise a spare pin. I021 is U0TXD; the ROM boot log will flicker the green emitter at reset. Accepted for the prototype: firmware logs via USB. J3 pin 1 is sense-only for the jig. The jig must NOT back-feed +3V3 into the regulator output: power from the cartridge. J3 sits on the board underside and is reached through the empty cartridge bay. It has no charging function.

Dropout check: TLV755P max dropout 238 mV at 500 mA (datasheet); scale roughly with load. ESP–NOW/Wi–Fi TX bursts can reach a few hundred mA, so cell IR + contact resistance + dropout set the usable cut-off. Firmware must stop recording at a gauge-reported threshold set from measured sag (bench item P1). C6 input, C7/C8 = C2/C3 output bulk. J2: 2 x VBAT + 2 x GND spring contacts land on pads on the cartridge lid. Cell protection (e.g. BQ2970) lives in the cartridge; charging, NTC and pack ID belong to the separate dock (not designed here).

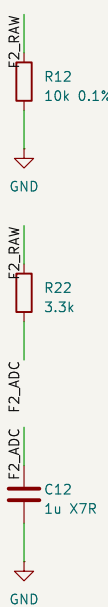
04 PASSIVE INSERT INTERFACE / FIVE DIVIDER CHANNELS (BENCH TOPOLOGY, NOT A201 REFERENCE CIRCUIT)



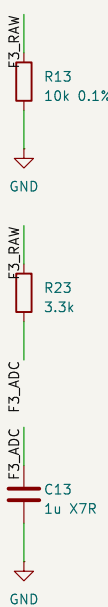
F1 (calibrate as built)



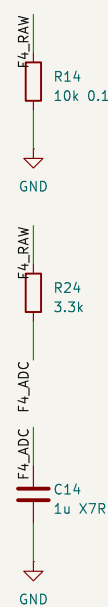
F2 (calibrate as built)



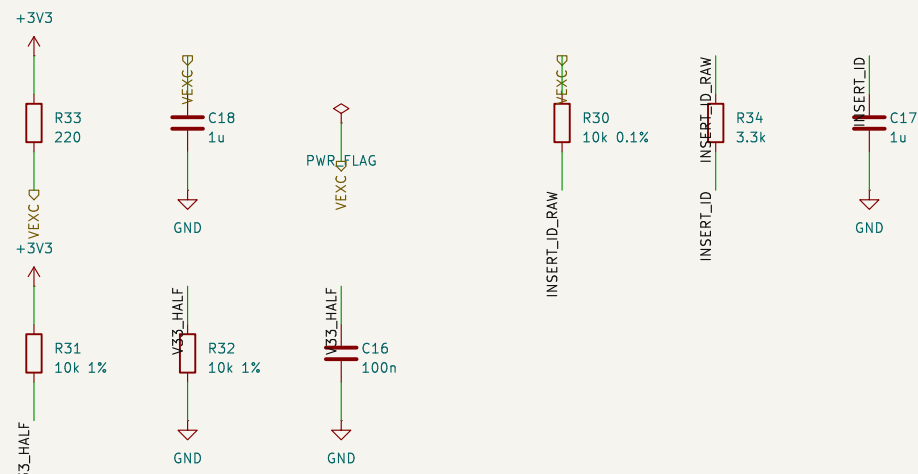
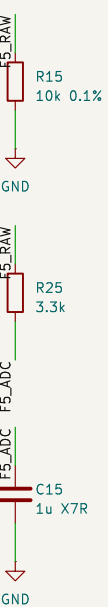
F3 (calibrate as built)



F4 (calibrate as built)



F5 (calibrate as built)



Force channel: FSR from VEXC to F_n_RAW, 10k to GND, 3.3k + 1 uF low-pass (f_c = 48 Hz) to the ADC. The 1 uF reservoir is ~50,000x the MCP3208 sample capacitor, so acquisition settling is set by the RC, not the ADC source-impedance limit. Excitation is continuous (Rev B decision kept): switching it every 5 ms cannot settle a 3.3 ms RC. R33 = 220R limits a tall short to 15 mA / 0.05 W (22R would dissipate 0.5 W in an 0603). VEXC sags to ~2.94 V at full load and moves with load, which cancels exactly because the ADC reference (pin 16) is the same VEXC node (MCP3208-REF = 160 uA, < 33 mV). ESP at J1: every force/ID line reaches the ADC through 3.3k into a 1 uF reservoir (0–10 mV IRM → 0–0.8 uC → ~0.8 V step). No TVS is fitted: the RC networks are the protection. System-level IEC 61000–4–2 testing is still required. Divider output is NOT the Tekscan A201 op–amp reference circuit and does not inherit its linearity or drift figures.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.

NIVA

Sheet: /Controller/
File: NIVA–controller.kicad_sch

Title: NIVA pod | Controller, acquisition, power, insert interface | Rev C

Size: A2 Date: 2026–09–28

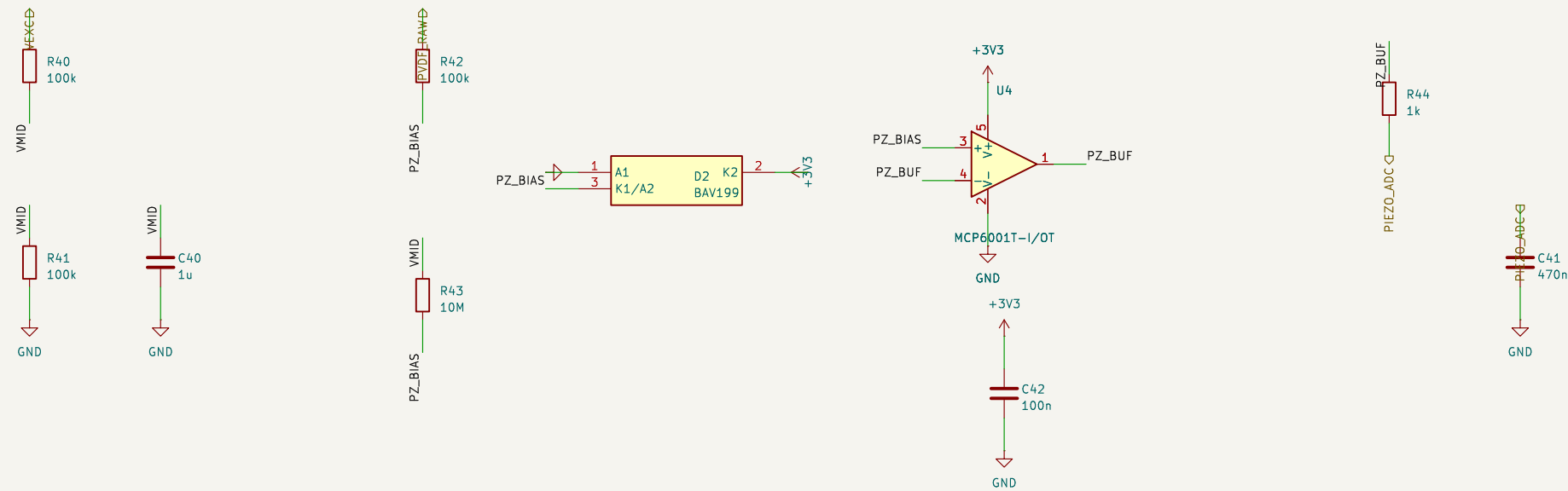
KiCad E.D.A. 9.0.9

Rev: C – ENGINEERING PROTO

Id: 2/3

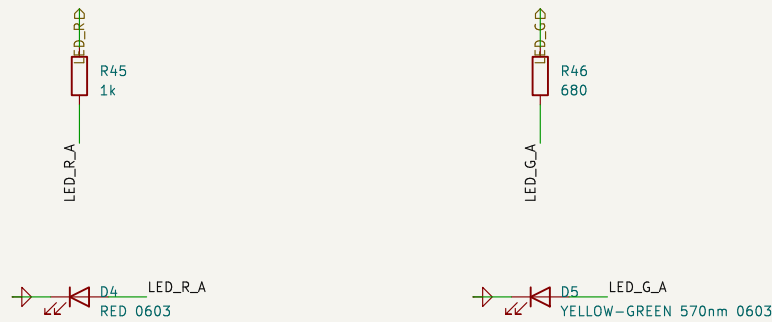
NIVA POD / HEEL PVDF BUFFER + STATUS / REV C ENGINEERING PROTOTYPE

01 MID-RAIL BIAS, INPUT PROTECTION, BUFFER



PVDF film (~1.4 nF class) with 10M bias: high-pass ~11 Hz. R42 limits clamp current during heel-strike spikes. D2 (BAV199 series pair: GND-A1, K1/A2-PZ_BIAS, K2-+3V3) clamps the buffer input to the rails; 3 pA typ leakage x 10M = 30 uV. R44/C41 = 339 Hz single pole for 2 kHz sampling; only ~10 dB at Nyquist. Treat as exploratory; oversample and decimate in firmware or add a 2nd-order stage after the E3 cable-motion experiment.

02 STATUS EMITTERS: RED + YELLOW-GREEN UNDER ONE LIGHT WINDOW



~1.3-1.7 mA per emitter at 3.3 V. Red + green together is read as amber. Light-window brightness and colour separation must be checked in the printed cover. Status colours signal recording quality only, never a clinical result.

Generated by work/build_niva_schematic_revC.py
NOT FOR FABRICATION. Values, protection and layout require bench validation.

NIVA

Sheet: /Heel and indicator/
File: NIVA-heel-and-indicator.kicad_sch

Title: NIVA pod | Heel vibration buffer and status emitters | Rev C

Size: A3 Date: 2026-09-28

KiCad E.D.A. 9.0.9

Rev: C - ENGINEERING PROTO
Id: 3/3